Cube-Sum Permutations: Graph Structure, Renormalization, and Exponential Families

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Abstract

Let $P(n)$ count permutations p of $[1, n]$ for which every $i + p(i)$ is an integer cube. We develop global, threshold-free structure: the cube-sum graph admits a partial matching of size $n - O(n^{2/3})$ for every n, and all Hall obstructions can be restricted to independent sets spanning the two boundary regions. The undirected cycle-cover expansion gives a parity determinant over $\mathbb{F}_2$, while a single valid matching factors exact counting through the strongly connected components of its alternation graph. An explicit four-reflection permutation on 171 elements yields the affine renormalization

$$P(343n + 171) \geq P(n)^{343},$$

and an entropy monotonicity theorem along its multiplicative orbits. Separately, an elementary weighted identity of cubes gives a fully algebraic exponential-subsequence construction

$$P(681707u^3) \geq 2^{148877u^3-1} \qquad (u \geq 1).$$

As a complementary finite classification, one explicit interval involution and 191 short augmenting words certify the all-n existence conjecture recorded in OEIS A006063; independent Hall and alternating-cycle certificates classify the related multiplicity sequence A096680. This last classification remains computer-assisted, unlike the global structural and algebraic constructions. All certificates and independent exact checkers accompany the manuscript; literature priority and external mathematical review remain open.

1 The reflection principle

Write $\mathcal{C} = \{k^3 : k = 2, 3, \dots\}$ and let $P(n)$ be the number of cube-sum permutations of $\{1, \dots, n\}$. By convention the empty permutation is permitted only in the general extension lemma; our classification concerns $n \geq 1$. Let $\mathcal{G} = \{n \geq 1 : P(n) > 0\}$.

Lemma 1 (Reflection and counting)**.** *If $0 \leq m < n$ and $c = m + n + 1$ is a cube, every valid permutation p on $\{1, \dots, m\}$ extends to a valid permutation q on $\{1, \dots, n\}$ via*

$$q_i = \begin{cases} p_i, & i \leq m, \\ c - i, & m < i \leq n. \end{cases}$$

For different admissible cubes c, the images of these extensions are disjoint. In particular,

$$P(n) \geq \sum_{\substack{c \in \mathcal{C} \\ n+1 \leq c \leq 2n}} P(c - n - 1), \tag{1}$$

with the convention $P(0) = 1$.

Proof. Because $c = m+n+1$, reflection $i \mapsto c-i$ permutes precisely the new interval $\{m+1, \dots, n\}$, and $i+q_i = c$ there. Old indices retain their cube sums. The embedding is injective for each c and two different c yield different last entries $q_n = c - n$. The claimed sum counts disjoint families. □

Lemma 2 (The cube-gap bound)**.** *For each $n \geq 365$ some cube c satisfies $n + 174 \leq c \leq 2n$. For each $n \geq 556$ some cube c satisfies $n + 176 \leq c \leq 2n$.*

Proof. For $365 \leq n \leq 555$, take $c = 729$. For $n \geq 556$, let k be the least integer with $k^3 \geq n + 176$; then $k \geq 10$ and $n \geq (k-1)^3 - 175$. Thus

$$2n - k^3 \geq k^3 - 6k^2 + 6k - 352 \geq 0.$$

The final polynomial equals 108 at $k = 10$ and has strictly positive successive differences $3k^2 - 9k + 1$ for $k \geq 10$. Hence the second assertion holds and implies the first for $n \geq 556$. □

2 Global graph structure, factorization, and parity

The cube-sum condition is symmetric in its two indices. This observation gives a conceptual interpretation of existence, multiplicity, exact counting, and parity that does not refer to any numerical threshold.

Let G_n be the undirected graph on $[1, n]$, with an edge $\{i, j\}$ whenever $i \neq j$ and $i + j$ is a perfect cube. At a vertex i, allow a loop exactly when $2i$ is a cube. A *spanning cycle-cover configuration* is a decomposition of its vertices into disjoint allowed loops, single edges (dimer components), and unoriented simple cycles of length at least three.

Lemma 3 (Arithmetic location of fixed points)**.** *In any cube-sum permutation, a fixed point $p(i) = i$ can occur only at*

$$\boxed{i = 4r^3 \qquad (r \geq 1).}$$

Consequently a valid involution of odd length must have at least one fixed point at such an index. In particular, a cube-sum permutation of odd length $n < 4$ cannot be an involution.

Proof. At a fixed point $2i = k^3$. The cube on the right is even, so $k = 2r$, whence $i = 4r^3$. An involution partitions its vertices into fixed points and transpositions; when n is odd at least one fixed point is necessary. □

Theorem 1 (Independent-set Hall criterion)**.** *A cube-sum permutation exists on $[1, n]$ if and only if every vertex set I of G_n containing no internal edge* or loop *satisfies*

$$\boxed{|N_{G_n}(I)| \geq |I|,} \tag{2}$$

where $N_{G_n}(I)$ includes all vertices adjacent to some vertex of I. Thus the full family of Hall inequalities can be reduced to independent *vertex sets in the symmetric cube-sum graph.*

Proof. The bipartite double cover of G_n has a perfect matching precisely when Hall's inequality $|N(S)| \geq |S|$ holds for every $S \subseteq [1, n]$. Its perfect matchings are exactly the cube-sum permutations. Hence the proposed condition is necessary.

Conversely, suppose Hall's inequality fails for some S. Put $I = S \setminus N(S)$. No two vertices of I can be adjacent, nor can I contain a loop: otherwise one of those vertices would lie in $N(S)$. Moreover $N(I) \cap S = \varnothing$, because any edge from I to S would again contradict $I \cap N(S) = \varnothing$. Thus

$$|N(I)| \leq |N(S) \setminus S| = |N(S)| - |N(S) \cap S| < |S| - |N(S) \cap S| = |I|.$$

So an independent Hall obstruction exists whenever any Hall obstruction exists. This proves sufficiency. □

In particular, every negative certificate supplied for $n < 173$ in the associated JSON is already an independent vertex set: none of its members are adjacent to each other or carry a permitted loop. The criterion is global and does not depend on any conjectural cutoff.

Theorem 2 (Uniform near-perfect partial matchings)**.** *For every integer $n \geq 7$, write*

$$k = \lfloor (n+1)^{1/3} \rfloor, \qquad g(k) = (k+1)^3 - k^3 = 3k^2 + 3k + 1.$$

There is an explicitly constructible cube-sum partial permutation *matching at least*

$$\boxed{n - \left\lfloor \frac{g(k)}{2} \right\rfloor} \tag{3}$$

indices to distinct values within $[1, n]$. Consequently the maximum matching of the cube-sum bipartite graph has deficiency $O(n^{2/3})$ for every *n, including lengths where no complete cube-sum permutation exists.*

Proof. Choose whichever of the consecutive cubes $c_- = k^3$ and $c_+ = (k+1)^3$ is closest to $n+1$, and denote it by c. Their gap is $g(k)$, hence $\delta = |c-(n+1)| \le \lfloor g(k)/2 \rfloor$. Since $n \ge 7$, $c_- \ge 8$. If $c_+ > 2n$, then $c_+ - (n+1) \ge n$ while $(n+1) - c_- \le n-7$, so the nearer cube must be c_-. Thus the chosen c lies in $[2, 2n]$.

The interval

$$J_c = [\max(1, c-n),\ \min(n, c-1)] \cap \mathbb{Z}$$

has precisely $n - \delta$ elements. It is invariant under $i \mapsto c - i$. On this interval set $p(i) = c - i$, giving a bijection $J_c \to J_c$ with the constant cube sum $i + p(i) = c$. Therefore it defines a partial perfect matching of size $n - \delta$. The claimed bound follows. □

Theorem 3 (Localization of Hall obstructions)**.** *For $n \ge 7$ let*

$$K = \lfloor (2n)^{1/3} \rfloor, \qquad G = 3K^2 + 3K + 1.$$

Every nonempty subset $S \subseteq [1, n]$ with diameter $\operatorname{diam}(S) = \max S - \min S$ *satisfying*

$$\boxed{\operatorname{diam}(S) \le n - G}$$

automatically obeys Hall's condition $|N(S)| \ge |S|$ in the symmetric cube-sum bipartite graph. Consequently every *independent Hall obstruction must satisfy*

$$\boxed{\operatorname{diam}(S) > n - G,}$$

and must contain vertices from both the lower and upper boundary regions of width at most G.

Proof. Write $a = \min S$, $b = \max S$. The candidate integer interval

$$[b+1, a+n]$$

has $n - (b-a) \ge G$ consecutive integers, all between 2 and $2n$. Within this range there cannot be G consecutive noncubes: the difference between consecutive cubes is $3j^2 + 3j + 1 \le G$ for $j \le K$, while the initial integers $2, \dots, 7$ also give fewer than G noncubes. Consequently some perfect cube c lies in $[b+1, a+n]$.

For every $i \in S$ we then have $1 \le c - i \le n$, and all $c - i$ are distinct. Thus $i \mapsto c - i$ is an injection of S into its neighborhood $N(S)$, proving Hall's condition. The contrapositive gives the claimed diameter restriction. The latter implies $\min S < G$ and $\max S > n - G + 1$, so a Hall obstruction has to meet both boundary regions. □

These theorems supply a completely noncomputational approach to the existence question. Instead of checking all subsets, one need only understand *long-spanning independent sets* whose vertices reach both ends of $[1, n]$. Eliminating those obstructions by a uniform argument would remove the remaining finite matching certificates; the near-perfect bound alone does not yet yield such an elimination.

Theorem 4 (Undirected cycle-cover expansion)**.** *If $\mathfrak{F}_n$ denotes the set of spanning cycle-cover configurations of G_n, and $c(F)$ is the number of its cycles of length at least three, then*

$$\boxed{P(n) = \sum_{F \in \mathfrak{F}_n} 2^{c(F)}.} \tag{4}$$

Thus $P(n) > 0$ if and only if G_n has a spanning union of loops, dimers, and simple cycles. In particular,

$$\boxed{P(n) \equiv I(n) \pmod 2,}$$

where $I(n)$ counts valid involutions.

Proof. Every valid permutation decomposes uniquely into fixed points, transpositions, and directed cycles of length at least three. The first correspond to permitted loops, the second to undirected dimer edges, and each longer undirected cycle admits exactly two orientations. Choices on different components are independent, giving (4). All terms with $c(F) > 0$ are even, whereas configurations with $c(F) = 0$ are exactly the involutions. □

Let H_n be the symmetric n-by-n zero-one matrix with $(H_n)_{ij} = 1$ precisely when $i + j$ is a perfect cube.

Corollary 1 (Efficient parity via binary algebra)**.**

$$\boxed{P(n) = \operatorname{per}(H_n), \qquad P(n) \bmod 2 = \det(H_n) \bmod 2.}$$

In particular $P(n)$ is odd exactly when H_n is nonsingular over $\mathbb{F}_2$, and its parity can be computed by polynomial-time Gaussian elimination without enumerating any permutations. Moreover,

$$\det(H_n) = 1 \pmod 2 \quad \Longrightarrow \quad \textit{there exists a legal involution}$$

since the number of such involutions is then odd. The converse need not hold.

Proof. The permanent expansion enumerates the admissible permutations. In characteristic two every permutation sign in the determinant is $+1$, so determinant and permanent coincide. □

Even more structure appears once any single legal permutation has been found. For such a permutation p, define the directed *alternation graph* D_p with vertex set $[1, n]$ and a non-loop arc $i \to j$ when $i + p(j)$ is a cube. Let $\mathcal{S}(D_p)$ be its strongly connected components, and let B_C be the square zero-one matrix indexed by $i, j \in C$, with $(B_C)_{ij} = 1$ if $i + p(j)$ is a cube (including all diagonal entries).

Theorem 5 (Exact strong-component factorization)**.** *For* every *legal reference permutation p of length n,*

$$\boxed{P(n) = \prod_{C \in \mathcal{S}(D_p)} \operatorname{per}(B_C).} \tag{5}$$

In particular $P(n) = 1$ precisely when every strong component is a singleton; and every nontrivial component contributes at least two possible permutations.

Proof. Any second legal permutation q determines a permutation $f = p^{-1} \circ q$ of $[1, n]$, and its arcs $i \to f(i)$ are allowed by the extended alternation graph with the identity loops included. Every cycle of f lies entirely inside one strongly connected component of D_p. Hence f restricts to independent permutations within the components, and their numbers multiply. Conversely, any such collection of internal permutation choices assembles into a legal q. The factorization follows. A nontrivial strong component contains a directed cycle, which can be toggled against the identity on that component, proving the final assertion. □

This theorem explains the previously computed $P(236) = 512$ from nine independent directed five-cycles and 191 singleton components. It also unifies the alternating-cycle criterion of Lemma 10 with exact counting. The companion verifier `global_graph_audit.py` checks the parity identity and component factorization in concrete, bounded-size instances using exact binary arithmetic.

3 Threshold-free renormalization and counting entropy

The specific all-n threshold is not needed for a powerful global closure property of the cube-sum permutation family.

Lemma 4 (Independent-fiber renormalization)**.** *Fix an integer exponent $d \geq 2$ and an odd positive integer t; set $L = t^d$, $A = (L-1)/2$, and $T_t(m) = Lm + A$. Let $P_d(m)$ denote the number of permutations of $[1, m]$ whose index-image sums are positive perfect dth powers. Then for every nonnegative integer m,*

$$\boxed{P_d(T_t(m)) \geq P_d(A) P_d(m)^L,} \tag{6}$$

with $P_d(0) = 1$. Moreover,

$$2T_t(m) + 1 = L(2m+1), \qquad T_s(T_t(m)) = T_{st}(m).$$

Proof. Partition $[1, T_t(m)]$ into a prefix $[1, A]$ and the remaining m consecutive blocks of length L. Choose an arbitrary valid permutation on the prefix, and choose independently L valid permutations $p_1, \ldots, p_L$ of $[1, m]$. At $x = A + (i-1)L + r$, $1 \leq r \leq L$, define

$$q(x) = A + (p_r(i) - 1)L + (L + 1 - r).$$

This permutes the remaining interval and satisfies

$$x + q(x) = L(i + p_r(i)).$$

Each right-hand side is a perfect dth power, since $L = t^d$. Different prefix permutations and different tuples of p_r produce different full permutations, yielding (6). The identities for T_t follow directly from its affine definition. □

Lemma 5 (A four-interval 171-permutation)**.** *The following involution is a cube-sum permutation of* $[1, 171]$*:*

$$p(i) = \begin{cases} 8 - i, & 1 \le i \le 7, \\ 27 - i, & 8 \le i \le 19, \\ 64 - i, & 20 \le i \le 44, \\ 216 - i, & 45 \le i \le 171. \end{cases}$$

Proof. The four displayed intervals are pairwise disjoint and partition $[1, 171]$. Each is invariant under its indicated reflection, whose index-image sum is, respectively, $2^3, 3^3, 4^3$, or 6^3. □

Theorem 6 (Cubic affine closure and entropy monotonicity)**.** *For every integer* $m \ge 1$*,*

$$\boxed{P(343m + 171) \ge P(m)^{343}.}$$

Set $T(m) = 343m + 171$*. Then for each positive integer* k*,*

$$T^k(m) = 343^k\Big(m + \frac{1}{2}\Big) - \frac{1}{2}, \qquad \boxed{P(T^k(m)) \ge P(m)^{343^k}.}$$

For $P(m) > 0$*, define the normalized entropy*

$$h(m) = \frac{\log P(m)}{m + \frac{1}{2}}.$$

It is nondecreasing on each orbit:

$$\boxed{h(T^k(m)) \ge h(m).}$$

Consequently, without requiring any eventual-existence theorem,

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \ge \sup_{\substack{m\ge 1 \\ P(m)>0}} \frac{\log P(m)}{m + \frac{1}{2}}.} \tag{7}$$

Proof. Apply Lemma 4 with $d = 3$, $t = 7$, $L = 343$, and $A = (343 - 1)/2 = 171$. Lemma 5 gives $P(171) \ge 1$. Iterating the inequality and the affine composition law yields the second display. Since $T^k(m) + 1/2 = 343^k(m + 1/2)$, the entropy inequality follows on taking logarithms. For each fixed m with $P(m) > 0$, the orbit $T^k(m)$ tends to infinity, so its entropy gives a lower bound for the global limsup. Taking the supremum over all such m proves (7), interpreted in the extended-real sense if necessary. □

The transformation $2m + 1 \mapsto 343(2m + 1)$ is a multiplicative renormalization of the odd coordinate $2m + 1$. In particular the set of admissible lengths is closed under $m \mapsto 343m + 171$, independently of the numerical threshold in Theorem 9. The explicit arithmetic verifier `renormalization.py` checks both the four-block seed and independent fiber choices without relying on the large finite-certificate archive.

4 A single-seed eventual existence theorem

There is a substantially more conceptual route to *eventual existence* than proving the sharp cutoff at 173. It uses only one seven-element permutation, seven rounds of integer reflection closure, and a uniform cubic gap inequality. In particular, this proof never invokes a bipartite matching algorithm or any of the earlier 173–364 permutation certificates.

Let $\mathcal{R}_0 = \{7\}$ and define a sequence of finite sets recursively by

$$\mathcal{R}_{j+1} = \mathcal{R}_j \cup \{k^3 - m - 1 : \ m \in \mathcal{R}_j, \ m < k^3 - m - 1 \le 10975, \ k \ge 2\}. \tag{8}$$

Lemma 6 (Seven-round arithmetic certificate)**.** *For $j = 0, \dots, 7$ the exact cardinalities $|\mathcal{R}_j|$ are*

$$1,\ 21,\ 189,\ 995,\ 3280,\ 6588,\ 8533,\ 8924.$$

Moreover,

$$\boxed{[8915, 10975] \cap \mathbb{Z} \subseteq \mathcal{R}_7.}$$

Proof. This is a bounded integer-only calculation, reproduced by `one_seed_eventual.py`. For each member m and each integer $k \geq 2$ with $k^3 \leq 21950$, form $n = k^3 - m - 1$ and insert it when $m < n \leq 10975$. After seven simultaneous rounds the cardinalities are exactly those displayed and all 2061 required lengths are present. The computation tests only elementary integer identities and finite set membership, not the existence of matchings. Independently, the verifier `one_seed_interval_proof.py` operates entirely on unions of integer intervals. For every current interval $[a, b]$ and cube c, it restricts to

$$\ell = \max(a, c - 10976), \qquad h = \min(b, \lfloor (c-2)/2 \rfloor)$$

and, if $\ell \leq h$, adds the reflected interval $[c - h - 1, c - \ell - 1]$. The exact numbers of disjoint intervals after rounds 0 through 7 are

$$1,\ 21,\ 186,\ 870,\ 2077,\ 2037,\ 1031,\ 770.$$

The last interval of the final union is exactly $[8915, 10975]$, which proves the finite coverage using *interval arithmetic alone*. Both verifiers reproduce the same seven-stage cardinalities. This finite check is separate from, and much weaker than, a search for perfect matchings. □

Theorem 7 (Eventual existence from one permutation)**.** *The explicit permutation*

$$\boxed{p = (7, 6, 5, 4, 3, 2, 1)}$$

at length 7, whose cube sums all equal 8, together with the reflection-extension lemma, proves that

$$\boxed{P(n) > 0 \quad \textit{for every } n \geq 8915.}$$

Thus eventual existence requires only one elementary starting permutation and the seven-round arithmetic certificate, without locating the optimal cutoff.

Proof. Each element of $\mathcal{R}_j$ is a length possessing a valid cube-sum permutation, by induction on j using Lemma 1. Therefore the finite interval $8915 \leq n \leq 10975$ is covered.

For $n \geq 10976$, let k be the least integer such that $k^3 \geq n + 8916$ and put $m = k^3 - n - 1$. Then $m \geq 8915$, and $k \geq 28$. We prove $k^3 \leq 2n$. If $k = 28$, this follows from $n \geq 10976$, since $28^3 = 21952$. If $k \geq 29$, minimality gives $n \geq (k-1)^3 - 8915$, whence

$$\begin{aligned} 2n - k^3 &\geq 2((k-1)^3 - 8915) - k^3 \\ &= k^3 - 6k^2 + 6k - 17832 \\ &\geq 1685 > 0. \end{aligned}$$

The last polynomial equals 1685 at $k = 29$ and is strictly increasing for all larger integer k. Thus $8915 \leq m < n$. By strong induction $P(m) > 0$, so reflection through the cube k^3 produces a valid permutation of length n. This proves the theorem. □

Corollary 2 (Efficient one-seed construction)**.** *For every $n \geq 8915$, a cube-sum permutation of length n is constructible using $O(n)$ unit-cost arithmetic operations and $O(\log \log n)$ reflection layers beyond a fixed finite lookup.*

Proof. For $n \geq 10976$, take the least cube $k^3 \geq n + 8916$ as in the theorem. Its predecessor satisfies

$$8915 \leq m = k^3 - n - 1 = O(n^{2/3}),$$

since consecutive cubes differ by $O(k^2) = O(n^{2/3})$. Hence $\log m \leq (2/3) \log n + O(1)$, which yields $O(\log \log n)$ recursive stages before reaching the finite interval. Every reflection writes its newly added positions exactly once, so the output work is $O(n)$; exact cube roots add only polylogarithmic overhead. The finite interval is traced through at most seven predecessor links, ending at the sole seed of length 7. □

Theorem 7 is weaker numerically than the fully classified cutoff in Theorem 9, but stronger conceptually in a different direction: it establishes eventual existence without any large explicit matching witness. The residual finite verification concerns *reflection reachability only*, not bipartite matching, and is entirely independent of the sharp-threshold classification. A completely computation-free proof of eventual existence would still require eliminating this last seven-round finite arithmetic check.

5 Monotone signed Waring representations

The reflection orbit has a direct reformulation as an additive-number-theoretic problem. The important feature is not just the presence of signed cubes, but the *monotonicity of every intermediate partial reflection.*

Proposition 1 (Reflection paths and alternating cubes)**.** *Fix any positive starting length a and set $x_0 = 2a+1$. A sequence of r strict reflection extensions from a to n exists if and only if there are integers $k_1, \dots, k_r \geq 2$ for which*

$$x_j = 2k_j^3 - x_{j-1} > x_{j-1} \qquad (1 \leq j \leq r),$$

and $x_r = 2n+1$. Equivalently,

$$\boxed{2n+1 = (-1)^r(2a+1) + 2\sum_{j=1}^{r}(-1)^{r-j}k_j^3,}$$

where the intermediate quantities specified by the recursion must be strictly increasing positive odd integers.

Proof. Writing $x_j = 2n_j + 1$, the reflection condition $n_{j-1} + n_j + 1 = k_j^3$ is equivalent to $x_{j-1} + x_j = 2k_j^3$. The strict extension $n_j > n_{j-1}$ is exactly $x_j > x_{j-1}$. Iterating this affine identity gives the alternating sum formula. Conversely every such increasing sequence recovers the necessary reflection predecessors and cubes. □

For the canonical seed $a = 7$, a path of exactly seven reflections reaches n precisely when it admits

$$\boxed{n + 8 = k_7^3 - k_6^3 + k_5^3 - k_4^3 + k_3^3 - k_2^3 + k_1^3}$$

with the corresponding strictly increasing intermediate odd coordinates. This converts the global question about *bounded reflection depth* into an explicit restricted signed-Waring representation problem. It is stronger than mere eventual existence: Theorem 7 proves a growing but $O(\log\log n)$-bounded depth, not a universal depth of seven.

Experiments with exact interval arithmetic suggest that the seven-step orbit already covers every length $n \geq 8915$, but this stronger unbounded statement is *not proved here.* It would require a uniform theorem about alternating cubic representations with their intermediate-order constraints, rather than an unbounded numerical extrapolation.

6 The reflection tree as a standalone counting model

The one-seed construction contains not merely one permutation for every sufficiently large length, but a recursively countable subfamily whose size itself grows uniformly.

Let $F(7) = 1$, and for $n > 7$ define

$$\boxed{F(n) = \sum_{\substack{7 \leq m < n \\ m+n+1 = k^3 \text{ for some } k \geq 2}} F(m).} \tag{9}$$

We regard $F(n)$ as zero at lengths not reachable from the seven-element seed.

Lemma 7 (Exact reflection-history enumeration)**.** *For every $n \geq 7$, $F(n)$ equals exactly the number of distinct cube-sum permutations obtainable from the seed $(7,6,5,4,3,2,1)$ by a finite sequence of strict reflection extensions. In particular,*

$$\boxed{0 \leq F(n) \leq P(n), \qquad F(n) > 0 \quad (n \geq 8915).}$$

Proof. Every strictly increasing history has a unique final predecessor $m < n$, and its last extension uses the cube $c = m + n + 1$. The final entry of the permutation is then

$$p_n = c - n = m + 1.$$

Thus the last predecessor m can be recovered uniquely from p_n, while the previous permutation is recovered by restricting to its first m positions. Hence two distinct reflection histories cannot produce the same final permutation. The recursive count is exactly (9). The eventual positivity is Theorem 7. □

Theorem 8 (Uniform entropy inside one reflection tree)**.** *Fix $s \geq 2$ and let $\rho_s \in (1, 2)$ be determined by $\sum_{h=1}^{s} \rho_s^{-h} = 1$. Put*

$$B_s^* = \max\{2s^2 + 4s - 1,\ 2s + 27\}, \qquad K(n) = \lfloor \sqrt[3]{2n} \rfloor.$$

For every $n \geq 8915$,

$$\boxed{F(n) \geq \rho_s^{(K(n) - B_s^*)/2}.}$$

Consequently,

$$\boxed{\liminf_{n\to\infty} \frac{\log F(n)}{(2n)^{1/3}} \geq \frac{\log 2}{2}.} \tag{10}$$

The same bound holds for $P(n)$, but this proof uses only the seven-element seed, its seven-round arithmetic reachability certificate, and elementary cubic inequalities.

Proof. Write $k = K(n)$, $\rho = \rho_s$, and $B = B_s^*$. If $k < B$, the right-hand side of the claimed inequality is at most 1, and $F(n) \geq 1$ by Lemma 7.

If $k \geq B$, for $j = 0, \ldots, s - 1$ take $c_j = (k - j)^3$ and $m_j = c_j - n - 1$. The polynomial inequality in the proof of Theorem 19 (applied with the same j, k) gives

$$K(m_j) \geq k - 2j - 2.$$

Since $k - 2s \geq 27$, this guarantees $m_j \geq \lceil 27^3/2 \rceil > 8915$. Also $c_j \leq k^3 \leq 2n$, so $m_j < n$. By strong induction, the exact recurrence (9) therefore yields

$$\begin{aligned} F(n) &\geq \sum_{j=0}^{s-1} F(m_j) \\ &\geq \rho^{(k-B)/2} \sum_{j=0}^{s-1} \rho^{-(j+1)} = \rho^{(k-B)/2}. \end{aligned}$$

Finally, for every fixed s, B_s^* does not depend on n, and $\rho_s \uparrow 2$ as $s \to \infty$. The limit inferior is at least $(\log \rho_s)/2$ for each s, which proves (10). □

The reflection-tree recurrence is a useful independent combinatorial object: its members are explicit permutations, not merely abstract paths or potential matchings. Unlike a claim that every large permutation can be generated by reflection (which is false, for example at $n = 173$ from this seed), this theorem counts a rigorously defined *subfamily* only.

7 A short, local existence certificate

The preceding reflection argument is infinite and elementary. The finite existence verification can itself be made much more transparent than an array of 192 long permutations: we give *one* explicit involution at size 173, then 191 short alternating-path words. Only 1844 single-digit cube roots are needed.

Lemma 8 (An explicit order-173 involution)**.** *There is a cube-sum permutation on* $[1, 173]$*, given on each listed interval* I *by* $p(i) = c - i$*. Intervals paired by a double arrow use the same cube* c*:*

intervals I	c
$[1, 7]$	8
$[22, 42]$	64
$[106, 110]$	216
$[8, 12] \longleftrightarrow [15, 19]$	27
$[13, 14] \longleftrightarrow [111, 112]$	125
$[20, 21] \longleftrightarrow [104, 105]$	125
$[43, 103] \longleftrightarrow [113, 173]$	216

Proof. The eleven intervals partition $[1, 173]$. Reflection $i \mapsto c - i$ preserves each of the first three intervals and exchanges the two intervals in each of the last four rows. Thus it defines an involution, and every sum $i + p(i)$ is a cube. □

Lemma 9 (An alternating-word extension)**.** *Suppose* p *is a cube-sum permutation of* $[1, n-1]$*. Let* $w = r_1 \cdots r_t$ *be a finite word of positive integer cube roots. Starting from* $u_0 = n$*, define* $v_j = r_j^3 - u_{j-1}$*. For* $j < t$*, require* $v_j \in [1, n-1]$ *and let* $u_j = p^{-1}(v_j)$*. Suppose all these* $u_0, u_1, \ldots, u_{t-1}$ *are distinct, all* $v_1, \ldots, v_t$ *are distinct, and* $v_t = n$*. Then replacing the old matching edges by*

$$q(u_{j-1}) = v_j \quad (1 \le j \le t),$$

while keeping all other old assignments, yields a cube-sum permutation of $[1, n]$*.*

Proof. The old matching edges removed along this path are $u_j \mapsto v_j$ for $j < t$; these form a bijection between the old visited left and right vertices. Their replacements join u_{j-1} to v_j and have sum r_j^3 by construction. The new left vertex n and the new right vertex n are incorporated, without repeating any endpoint. Every untouched vertex retains its old matching edge. □

The `augmenting_cube_words.txt` appendix lists one word for each $n = 174, \ldots, 364$. Every root is a single decimal digit in $\{2, \ldots, 8\}$. The words are checked using Lemma 9 and the predecessor matching obtained from the previous word; this is *local* verification, not a matching search or a scan to a large cutoff. The independent, standard-library-only `verify_word_paths.py` checks all required edge equalities and endpoint distinctness. The seven-line interval seed plus 1844 root digits certifies all 192 base cases of Theorem 9; combined with the cube-gap induction this proves existence for *every* $n \ge 173$. The older long witness archives remain as secondary independent cross-checks, not as prerequisites of this compact proof.

8 Finite certificates and exact membership

Our finite data are provided as UTF-8 JSON arrays of integers. A valid witness p is checked solely by

$$\mathrm{sort}(p) = (1, \ldots, n), \qquad \forall i \quad i + p_i \in \mathcal{C}.$$

For a negative instance, let $N_n(S) = \{j \in [1, n] : \exists i \in S,\ i + j \in \mathcal{C}\}$. Hall's marriage theorem gives the immediate obstruction

$$|N_n(S)| < |S| \quad \Longrightarrow \quad P(n) = 0. \tag{11}$$

The file `exact_classification_below_173.json` provides 48 explicit valid permutations and 124 explicit Hall-deficient subsets, exactly partitioning $1 \le n \le 172$. Every assertion is directly checked by `verify_extended.py`. For instance, for $n = 172$ the 27 left indices

$$S = \{1, 5, 8, 13, 20, 31, 35, 38, 42, 43, 45, 92, 96, 99,$$
$$104, 111, 122, 126, 129, 133, 134, 136, 153, 157, 160, 165, 172\}$$

have precisely 26 distinct possible right neighbors. Thus $172 \notin \mathcal{G}$, making the threshold below sharp.

For the unbounded tail, only the 109 witness sizes

$$173 \le n \le 255, \qquad 339 \le n \le 364 \tag{12}$$

require explicit stored permutations. Indeed for each $256 \le n \le 338$, put $m = 511 - n \in [173, 255]$; then $m + n + 1 = 512 = 8^3$, so Lemma 1 constructs the missing 83 cases. This compresses the previously saved 192 base cases by $83/192 > 43\%$. All 109 stored witnesses in `core_109.json`, and all 192 reconstructed witnesses, are checked with exact integer arithmetic. The earlier full 192-row file is retained as an additional independent-format audit source.

There is a further useful compression when the 48 verified positive sizes below 173 are reused. An ascending reflection-closure algorithm starts from those small solutions and, at each $173 \le n \le 364$, uses any already certified $m < n$ for which $m+n+1$ is a cube. Precisely 132 lengths are obtained by reflection and only 60 further lengths need explicit witnesses. Here the empty matching is allowed: $n = 342$ follows directly from $c = 343$, with $p(i) = 343 - i$. The 60 additional witnesses are archived in `core_60_with_empty.json`, and `verify_extended.py` certifies this alternate compression independently of the 109-case presentation.

Proposition 2 (Limit of reflection-only compression)**.** *Suppose the empty matching and the* 48 *known positive lengths below* 173 *are provided as initial data. Among proofs of all lengths* $173, \dots, 364$ *that use only the reflection extension of Lemma 1 and independently supplied new full-length seed permutations, at least* 60 *extra seeds are necessary, and* 60 *suffice.*

Proof. Process $n = 173, \dots, 364$ in ascending order. If no cube c yields a previously certified positive predecessor $m = c - n - 1 \in [0, n-1]$, then reflection cannot establish n from earlier data, so a new seed is forced. Otherwise choose such a predecessor and perform reflection, without adding a seed. This deterministic closure rule yields 44 forced seeds in $[173, 255]$, none in $[256, 338]$ (use $c = 512$ and $m = 511 - n$), and 16 in $[339, 364]$. Thus $44 + 16 = 60$, and the attached short integer-only checker reproduces the forced index list. Every forced seed is unavoidable by the definition of the restricted proof rule. □

This is *not* a lower bound on arbitrary mathematical proofs. The alternating-path extension in Lemma 9 is a different operation and indeed replaces those 60 large independent seeds by one explicit involution and a small word table.

Theorem 9 (Complete A006063 membership)**.** *For every integer* $n \ge 173$*,* $P(n) > 0$*. Below* 173*, the complete membership list is*

$$\begin{aligned}\mathcal{G} \cap [1, 172] = \{&7, 19, 26, 37, 44, 56, 63, 66, 68, 80, 82, 85, 87, 98, 100,\\ &103, 105, 110, 112, 115, 116, 117, 118, 119, 120, 121,\\ &122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132,\\ &133, 135, 147, 149, 150, 151, 152, 155, 156, 159, 171\}.\end{aligned}$$

In sequence-index notation this gives

$$\boxed{\text{A006063}(j) = j + 124 \quad (j \ge 49).}$$

Proof. The exact finite witnesses establish $173 \le n \le 364$, and the Hall certificates settle the smaller sizes. For $n \ge 365$, Lemma 2 gives a cube $c \in [n + 174, 2n]$. The predecessor $m = c - n - 1$ satisfies $173 \le m < n$. By strong induction $P(m) > 0$, and Lemma 1 yields $P(n) > 0$. There are precisely 48 positive indices below 173, so the 49th and all later sequence terms are consecutive. □

9 Multiplicity and the second OEIS classification

A cube-sum permutation p is a perfect matching in the bipartite graph with left and right vertices $1, \dots, n$, where (i, j) is an edge exactly when $i + j \in \mathcal{C}$. For any matching p, form a directed graph D_p on the left vertices, putting $i \to j$ whenever $i \ne j$ and (i, p_j) is a cube-sum edge.

Lemma 10 (Alternating-cycle criterion)**.** *For a valid permutation* p*, there exists another distinct valid permutation if and only if* D_p *has a directed cycle.*

Proof. Along a directed cycle $i_1 \to \cdots \to i_t \to i_1$, replace p_{i_s} by $p_{i_{s+1}}$ (cyclic indices). Every moved pair is an allowed edge, and all right vertices remain distinct. Conversely, if $q \ne p$ is another valid permutation, the nontrivial cycles of the permutation $p^{-1} \circ q$ consist of directed cycles in D_p. Thus acyclicity is a rigorous uniqueness certificate. □

The file `second_solutions.json` records directed cycles, not complete duplicate permutations. Direct finite checking establishes: among the 48 positive lengths below 173, exactly

$$E = \{112, 115, 116, 117, 119, 124, 125, 126, 127, 128, 129, 130, 133\}$$

have a directed alternating cycle; every other positive length below 173, as well as 173 and 174, has an acyclic D_p and therefore a unique permutation.

Furthermore, all 190 base cases $175 \le n \le 364$ have *two distinct* alternating-cycle certificates, archived as `second_solutions.json` and `third_solutions.json`. They yield three different permutations, counting the original matching. To extend the multiplicity assertion: for $365 \le n \le 553$, the cube 729 gives $m = 728 - n \in [175, 363]$. The two exceptional lengths $554, 555$ have separate pairs of alternating-cycle certificates obtained by extending the unique solutions at $174, 173$ respectively. For $n \ge 556$, the stronger part of Lemma 2 gives $m = c - n - 1 \ge 175$. Three distinct solutions for m remain distinct after reflection, finishing an induction.

Theorem 10 (Complete A096680 membership)**.** *The positive integers with at least two cube-sum permutations are exactly*

$$\boxed{E \ \cup \ \{175, 176, 177, \dots\}.}$$

Moreover $P(n) \ge 3$ for every $n \ge 175$, while $P(173) = P(174) = 1$. The sequence terms satisfy

$$\boxed{\text{A096680}(j) = j + 161 \quad (j \ge 14).}$$

Sharpness of the constant three. The bound $P(n) \ge 3$ for all $n \ge 175$ cannot be improved to four: $P(175) = 3$ exactly. For the stored matching at 175, the directed alternation graph has precisely one nontrivial strongly connected component,

$$\{4, 11, 16, 23, 41, 48, 102, 114, 168, 175\},$$

and the other 165 vertices are singleton components. Every cycle of a permutation alternative lies in a strongly connected component, so the permanent reduces to the 10-by-10 component's allowed-edge matrix (including the matching self-edges). A direct bitmask dynamic program gives its permanent as 3. This computation is independently reproduced in `verify_extended.py` using exact integers.

10 A nine-gadget exceptional case at $n = 236$

An *involution* is a permutation satisfying $p^2 = \mathrm{id}$. At $n = 236$ there are exactly 512 cube-sum permutations, *none* of which is an involution. This follows from a short alternating-cycle decomposition, not from a general permanent computation.

Define a reference permutation p of $[1, 236]$ by $p(i) = c - i$ on the following fifteen intervals:

I	c	I	c
$[1, 7]$	8	$[8, 9]$	216
$[10, 18]$	64	$[19, 43]$	125
$[44, 45]$	216	$[46, 70]$	125
$[71, 81]$	216	$[82, 106]$	125
$[107, 134]$	343	$[135, 136]$	216
$[137, 170]$	343	$[171, 172]$	216
$[173, 197]$	343	$[198, 208]$	216
$[209, 236]$	343		

These interval images partition $[1, 236]$, and every prescribed sum is a cube.

Lemma 11 (Translation of alternating cycles)**.** *Suppose $p(i) = c_j - i$ on pairwise disjoint integer intervals I_j for $1 \le j \le r$. If $i_j \in I_j$ form a directed alternating cycle $i_1 \to i_2 \to \cdots \to i_r \to i_1$ for p, then every integer shift t with $i_j + t \in I_j$ for all j produces another alternating cycle on $i_j + t$. If t ranges through a set T of such shifts, the translated cycles are disjoint and yield at least $2^{|T|}$ distinct valid cube-sum permutations.*

Proof. Because $p(i_{j+1}+t) = c_{j+1} - i_{j+1} - t$,

$$(i_j + t) + p(i_{j+1} + t) = i_j + c_{j+1} - i_{j+1} = i_j + p(i_{j+1}),$$

so every translated alternating edge has the same cube sum as its original. Since the I_j are pairwise disjoint, different shifts cannot share a vertex within any I_j. The cycles are therefore vertex-disjoint and can be switched independently. □

In the present 236-element template,

$$10 \to 137 \to 46 \to 198 \to 71 \to 10$$

is one alternating 5-cycle, with successive cube sums

$$216, \quad 216, \quad 64, \quad 343, \quad 125.$$

The five vertices belong respectively to the five different reflection blocks $[10, 18]$, $[137, 170]$, $[46, 70]$, $[198, 208]$, $[71, 81]$. The common shifts $t = 0, \ldots, 8$ produce nine disjoint alternating cycles *by pure algebra*, giving immediately

$$\boxed{P(236) \geq 2^9 = 512}$$

without any matching enumeration. The remaining work for the exact count is only to exclude further cycles.

Lemma 12 (Alternating digraph at 236)**.** *Let D_p be the digraph with an arrow $i \to j$ whenever $i \neq j$ and $i + p(j)$ is a cube. Its strongly connected components comprise* 191 *singletons and exactly nine five-vertex sets*

$$S_t = \{10 + t, 46 + t, 71 + t, 137 + t, 198 + t\}, \qquad 0 \leq t \leq 8.$$

Within each S_t the nontrivial arrows form one simple directed 5*-cycle, without chords.*

Proof. This is a finite integer check using the only six possible cubes $8, 27, 64, 125, 216, 343$. The interval formula specifies every edge; the independent `count_236.py` verifier checks all 236 rows, the component partition, and the five-cycle properties with integer arithmetic. It invokes neither a matching solver nor a permanent-computation routine. □

Theorem 11 (Exact count and involution obstruction)**.**

$$\boxed{P(236) = 2^9 = 512.}$$

There is no cube-sum involution on $[1, 236]$.

Proof. Given any other cube-sum permutation q, $f = p^{-1} \circ q$ is a permutation whose arrows lie in D_p or are identity loops. Every cycle of f is confined to one strongly connected component. Each singleton is therefore fixed, while each S_t has exactly two choices: five fixed points or the unique directed 5-cycle. The choices are independent, so there are 2^9 permutations q.

For the central set $S_4 = \{14, 50, 75, 141, 202\}$, the reference permutation restricts to

$$14 \mapsto 50 \mapsto 75 \mapsto 141 \mapsto 202 \mapsto 14.$$

The only two possible restrictions of q to S_4 are this 5-cycle and its inverse, neither of which is an involution. Thus no q is an involution. □

The failure of involutivity at 236 is therefore a specific five-cycle obstruction, not a failure of cube-sum permutation existence. The proof uses only a fifteen-row interval template and an easily audited finite graph-decomposition lemma.

11 A computation-free exponential construction

A particularly simple weighted identity of four cubes produces more permutations than all the previous asymptotic constructions, with *no* finite matching certificate or eventual-existence theorem.

Proposition 3 (Weighted-cube odd-cycle principle)**.** *Let $q \geq 1$, and let A, B, X, Y be positive cubes such that*

$$(q+1)X + qY = A + 2qB.$$

For $0 \leq j \leq 2q$ put

$$D_j = \lceil j/2 \rceil (B - X) + \lfloor j/2 \rfloor (B - Y).$$

Assume $A \leq D_j \leq B - 2A$ for $j \geq 1$ and $|D_j - D_k| \geq A$ whenever $j \neq k$. Then for every integer $u \geq 1$,

$$\boxed{P((B-A)u^3) \geq 2^{Au^3-1}.}$$

In fact the family encodes $Au^3 - 1$ independent binary choices, each of which changes precisely $2q+1$ permutation positions.

Proof. Put $T = u^3$ and define the two-interval reflection $p(i) = AT - i$ for $i < AT$ and $p(i) = BT - i$ for $AT \leq i \leq (B-A)T$. The interval hypotheses make this a valid permutation with cube sums AT and BT. For each $1 \leq s < AT$, let $v_j = s + D_j T$ for $0 \leq j \leq 2q$. The gap conditions place v_0 in the first reflection block and the other vertices in the second, while ensuring that the translated cycles for different s are disjoint. The edge sums $v_j + p(v_{j+1})$ are alternately XT and YT for $j < 2q$. The final edge is a cube as well, since

$$v_{2q} + p(v_0) = (A + 2qB - qX - qY)T = XT.$$

Thus the $AT - 1$ cycles can be switched independently, proving the count and Hamming-distance statement. □

For $q = 1$ this recovers the three-cycle identity $2X + Y = A + 2B$. For $q = 2$ it becomes $3X + 2Y = A + 4B$. In particular,

$$3 \cdot 5^3 + 2 \cdot 8^3 = 3^3 + 4 \cdot 7^3$$

gives $(A, B, X, Y) = (27, 343, 125, 512)$ and offsets $(D_0, \ldots, D_4) = (0, 218, 49, 267, 98)$, satisfying every hypothesis. Consequently

$$\boxed{P(316u^3) \geq 2^{27u^3-1}}$$

for every positive integer u. This unifies the three- and five-cycle constructions through one algebraic identity.

Lemma 13 (Three-cycle identity gadget)**.** *Let A, B, X, Y be positive perfect cubes satisfying*

$$A + 2B = 2X + Y, \qquad A + \tfrac{1}{2}B \leq X \leq B - A.$$

For every positive integer u, put $T = u^3$ and $N = (B-A)T$. There are at least

$$\boxed{2^{AT-1}}$$

different permutations p of $[1, N]$ for which $i + p(i)$ is a perfect cube.

Proof. The displayed inequalities give $B \geq 4A$. Define a reference permutation by two reflections:

$$p(i) = \begin{cases} AT - i, & 1 \leq i < AT, \\ BT - i, & AT \leq i \leq (B-A)T. \end{cases}$$

The first interval maps to itself under the first reflection, and the second also maps to itself under the second reflection. Every original sum is AT or BT, both perfect cubes.

Set $d_1 = B - X$ and $d_2 = X - A$. The hypotheses imply

$$d_1 \geq A, \qquad d_2 - d_1 \geq A, \qquad d_2 + A = X \leq B - A.$$

For every $s = 1, \ldots, AT - 1$, the triple

$$C_s = (s,\ s + d_1T,\ s + d_2T)$$

lies inside $[1, N]$, with the second and third vertices in the second reflection interval. The three translated ranges are pairwise disjoint by the displayed gap inequalities. Thus all $AT - 1$ triples are pairwise vertex-disjoint.

Along the directed three-cycle $s \to s + d_1T \to s + d_2T \to s$, the three alternative matching sums are

$$\begin{aligned} s + p(s + d_1T) &= XT, \\ (s + d_1T) + p(s + d_2T) &= (A + 2B - 2X)T = YT, \\ (s + d_2T) + p(s) &= XT. \end{aligned}$$

Because T and A, B, X, Y are cubes, every new sum is a cube. Switching any subset of the $AT - 1$ disjoint cycles yields 2^{AT-1} pairwise different legal permutations. □

Theorem 12 (Explicit high-entropy cube-sum permutations)**.** *For every integer $u \geq 1$,*

$$\boxed{P(681707u^3) \geq 2^{148877u^3-1}.} \tag{13}$$

Consequently,

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \geq \frac{148877 \log 2}{681707} > 0.} \tag{14}$$

This theorem is entirely computation-free, apart from elementary multiplication of four displayed integers.

Proof. Apply Lemma 13 with

$$\begin{aligned} A = 53^3 = 148877, &\quad B = 94^3 = 830584, \\ X = 87^3 = 658503, &\quad Y = 79^3 = 493039. \end{aligned}$$

Indeed,

$$A + 2B = 1810045 = 2X + Y,$$

and

$$A + \tfrac{1}{2}B = 564169 \leq X = 658503 \leq 681707 = B - A.$$

Hence the $AT - 1 = 148877u^3 - 1$ explicit cycles are disjoint. More precisely, each has the form

$$s \longrightarrow s + 172081u^3 \longrightarrow s + 509626u^3 \longrightarrow s,$$

with alternative sums $(87u)^3, (79u)^3, (87u)^3$. No matching search, graph decomposition, or auxiliary finite base case is involved. Taking logarithms along the sequence $n = 681707u^3$ proves the second claim. □

Corollary 3 (Linear-distance binary permutation codes)**.** *For $N = 681707u^3$ there is an injective map from $\{0,1\}^{148877u^3-1}$ into legal cube-sum permutations of $[1, N]$ that multiplies all Hamming distances by exactly three. Its distance enumerator is*

$$\left(1 + z^3\right)^{148877u^3-1}.$$

In particular, a greedy binary code subfamily has exponentially many legal permutations with pairwise Hamming distance proportional to N.

Proof. Each bit chooses whether to switch one of the pairwise disjoint three-cycles. A changed bit changes precisely three mapped positions. The distance identity and polynomial follow immediately. The greedy Hamming-ball argument from Corollary 5 applies unchanged. □

12 Power-block lifting and exponential growth on a subsequence

The nine disjoint alternating cycles at 236 can be amplified by a purely algebraic interval construction, yielding exponentially many solutions at infinitely many *different* lengths. The lifting mechanism works for every perfect-power exponent, not only cubes.

Lemma 14 (Independent-fiber perfect-power tensorization)**.** *Fix $d \geq 2$ and $M \geq 1$, and suppose $P_d(M) > 0$. For any odd positive integer t put $L = t^d$ and $A = (L-1)/2$. For each $r \in [1, L]$ choose independently a dth-power-sum permutation p_r of $[1, M]$. Define a permutation Q on the interval $I = [A+1, A+ML]$ by*

$$Q\big(A + (i-1)L + r\big) = A + (p_r(i) - 1)L + (L + 1 - r). \tag{15}$$

Then every $x + Q(x)$ is a perfect dth power, and the L independent choices give exactly

$$\boxed{P_d(M)^L}$$

different permutations of I.

If instead all fibers use the same p possessing h pairwise vertex-disjoint directed alternating cycles, then Q has at least hL pairwise vertex-disjoint directed alternating cycles.

Proof. The input positions have unique coordinates (i, r). For fixed r, the map $i \mapsto p_r(i)$ is a permutation of $[1, M]$, while the output residue $L + 1 - r$ identifies the original fiber r uniquely. Thus (15) is a bijection. For $x = A + (i-1)L + r$,

$$\begin{aligned} x + Q(x) &= 2A + L(i + p_r(i) - 1) + 1 \\ &= L(i + p_r(i)). \end{aligned}$$

If $i + p_r(i) = k_{i,r}^d$, then this sum equals $(tk_{i,r})^d$. The map from the tuple $(p_1, \ldots, p_L)$ to Q is injective, since all $p_r(i)$ are recovered from the output block indices; hence there are exactly $P_d(M)^L$ such interval permutations.

Finally, when every $p_r = p$, an alternating edge $i \to j$ of p gives $x_{i,r} + Q(x_{j,r}) = L(i + p(j))$ for each $x_{i,r} = A + (i-1)L + r$, again a perfect power. Every original alternating cycle consequently lifts to one cycle in each fiber, and disjoint fibers and original cycles remain disjoint. □

Theorem 13 (An exponential subsequence)**.** *For every odd integer $t \geq 9$, define*

$$\boxed{n_t = \frac{551t^3 - 1}{2}.}$$

Then

$$\boxed{P(n_t) \geq P\left(\frac{t^3 - 1}{2}\right) 512^{t^3} \geq 3 \cdot 2^{9t^3}.}$$

In particular,

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \geq \frac{18 \log 2}{551} > 0.}$$

Proof. Take the explicit $M = 236$ permutation from Section 10, possessing $h = 9$ disjoint alternating 5-cycles. Let $L = t^3$, $A = (L-1)/2$, and $B = A + 236L = (473L - 1)/2$. Because $t \geq 9$, $A \geq 364 \geq 173$, so Theorem 9 supplies a valid permutation on the prefix $[1, A]$. By Theorem 10, $P(A) \geq 3$. The exact count $P(236) = 512$ from Theorem 11 and Lemma 14 provide 512^L distinct valid middle-block permutations on $[A+1, B]$, without needing to select a preferred alternating-cycle orientation.

Now let $C = (8t)^3 = 512L$ and set

$$n_t = C - B - 1 = \frac{551L - 1}{2}.$$

The final suffix $[B+1, n_t]$ has exactly $39L > 0$ elements. Define the permutation there by $x \mapsto C - x$. It permutes that suffix because $C = B + n_t + 1$, and every new sum is the cube C. Combining the three disjoint intervals produces a cube-sum permutation on $[1, n_t]$. The prefix and middle-block choices are independent, while the suffix is fixed. Consequently at least $P(A)\, 512^L$ distinct permutations are obtained, and $P(A) \geq 3$ completes the proof. The older $9L$ independent-cycle construction is a special subfamily of the new tensorized construction. □

Theorem 14 (Asymptotically optimal terminal reflection)**.** *Let $t \geq 9$ be odd and $L = t^3$. Let k_t be the least integer satisfying*

$$k_t^3 > 473L,$$

and define

$$\boxed{N_t = k_t^3 - \frac{473L+1}{2}.}$$

Then

$$\boxed{P(N_t) \geq P\left(\frac{L-1}{2}\right) 512^L \geq 3 \cdot 2^{9L}.}$$

Furthermore,

$$N_t = \left(\frac{473}{2} + o(1)\right) L \qquad (t \to \infty),$$

and therefore

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \geq \frac{18 \log 2}{473}.} \tag{16}$$

Proof. Take the 236-element tensorized gadget of Lemma 14. The valid prefix has length $A = (L-1)/2$ and the lifted middle block ends at $B = (473L-1)/2$. The strict inequality $k_t^3 > 473L = 2B+1$ guarantees the suffix interval $[B+1, N_t]$ is nonempty: indeed

$$N_t - B = k_t^3 - 473L > 0.$$

Reflection through the cube k_t^3 permutes that suffix since $B + N_t + 1 = k_t^3$. The middle blocks have 512^L independent choices, the prefix has $P(A)$ choices, and the suffix is fixed. The first bound follows, and $P(A) \geq 3$ because $A \geq 364$. By minimality,

$$(k_t - 1)^3 \leq 473t^3 < k_t^3,$$

so $k_t = 473^{1/3}t + O(1)$. Thus $k_t^3 = 473t^3 + O(t^2)$ and $N_t = (473/2)t^3 + O(t^2)$. Dividing $9L \log 2$ by N_t establishes (16). □

The fixed terminal cube $(8t)^3$ used in Theorem 13 is convenient for a simple closed formula $n_t = (551t^3 - 1)/2$, but is not asymptotically optimal. Selecting the *least* admissible terminal cube as above shrinks the suffix from $39t^3$ to $O(t^2)$, and improves the linear entropy coefficient from $18 \log 2/551$ to $18 \log 2/473$.

Corollary 4 (Multiplicity forces exponential subsequences)**.** *Fix $d \geq 2$. Suppose $P_d(n) > 0$ for all sufficiently large n and that $P_d(M) \geq 2$ for some $M \geq 1$. Choose any integer b with $b^d > 2M+1$. For all sufficiently large odd t, define*

$$n_t = \left(b^d - M - \tfrac{1}{2}\right)t^d - \tfrac{1}{2}.$$

Then

$$P_d(n_t) \geq P_d\left(\frac{t^d-1}{2}\right) P_d(M)^{t^d},$$

and in particular

$$\boxed{\limsup_{n\to\infty} \frac{\log P_d(n)}{n} \geq \frac{\log P_d(M)}{b^d - M - \frac{1}{2}} > 0.}$$

Proof. Let $L = t^d$, $A = (L-1)/2$ and $B = A + ML$. Eventual existence supplies a valid prefix on $[1, A]$ for all sufficiently large odd t, with $P_d(A)$ possible choices. Lemma 14 gives $P_d(M)^L$ independent middle-block choices. Now put $C = (bt)^d = b^d L$ and $n_t = C - B - 1$. Since $b^d > 2M+1$, the suffix $[B+1, n_t]$ is nonempty; reflection $x \mapsto C - x$ permutes it and makes every sum equal to C. The three intervals are invariant under the constructed permutation, and their choices are independent. Taking logarithms and dividing by n_t gives the bound. □

Theorem 15 (An exact binary cube-sum permutation code)**.** *For each odd $t \geq 9$, put $L = t^3$ and $n_t = (551L - 1)/2$. There is an injective map*

$$\Phi_t : \{0,1\}^{9L} \longrightarrow \{\textit{cube-sum permutations of } [1, n_t]\}$$

such that for all bit strings u, v,

$$\boxed{d_{\mathrm{H}}(\Phi_t(u), \Phi_t(v)) = 5\, d_{\mathrm{H}}(u, v).} \tag{17}$$

In particular, relative to $\Phi_t(0)$, the exact Hamming-distance enumerator of this explicit family is

$$\boxed{\sum_{u \in \{0,1\}^{9L}} z^{d_{\mathrm{H}}(\Phi_t(u), \Phi_t(0))} = (1 + z^5)^{9L}.}$$

Proof. The proof of Theorem 11 identifies every 236-element cube-sum permutation with a unique nine-bit vector recording which of the nine disjoint directed five-cycles have been switched. Two different choices on one five-cycle change all five mapped values, while all untouched components agree. Thus their permutation Hamming distance is exactly five times the bitwise Hamming distance.

Take the L independent base-permutation fibers of Lemma 14, and concatenate their nine-bit selectors into $u \in \{0,1\}^{9L}$. Different fibers occupy disjoint input positions, so distances add exactly. Finally append a fixed valid prefix and the fixed cube-reflection suffix of Theorem 13. Neither changes the Hamming distances. The generating polynomial follows because each of the $9L$ bits independently contributes either distance zero or distance five. □

Corollary 5 (Positive-rate, linear-distance families)**.** *Let $H_2(x) = -x \log_2 x - (1-x) \log_2 (1-x)$ be binary entropy. For every odd $t \geq 9$, there is a subfamily $\mathcal{F}_t$ of cube-sum permutations on $[1, n_t]$ such that*

$$|\mathcal{F}_t| \geq 2^{\,9L(1 - H_2(1/4))}, \qquad d_{\min}(\mathcal{F}_t) > \frac{45L}{4}.$$

Consequently both $\log |\mathcal{F}_t| = \Omega(n_t)$ and $d_{\min}(\mathcal{F}_t) = \Omega(n_t)$.

Proof. Greedily select a binary word of length $N = 9L$, and delete all words within Hamming distance $\lfloor N/4 \rfloor$ of the selected word. A binary Hamming ball of this radius has at most $2^{N H_2(1/4)}$ members, so the greedy packing yields at least $2^{N(1 - H_2(1/4))}$ codewords, with pairwise binary distance exceeding $N/4$. Applying Φ_t multiplies all distances by exactly five, while preserving cardinality. Since $n_t = (551L - 1)/2$, both claimed linear asymptotics follow. □

Corollary 6 (Optimal general terminal choice)**.** *Under the assumptions of Corollary 4, the stronger universal estimate holds:*

$$\boxed{\limsup_{n \to \infty} \frac{\log P_d(n)}{n} \geq \frac{\log P_d(M)}{M + \frac{1}{2}}.}$$

Proof. Put $L = t^d$ with odd t and $B = ((2M+1)L - 1)/2$. Take the least integer k for which $k^d > (2M+1)L$. Since $k = (2M+1)^{1/d} t + O(1)$, the reflected suffix gives a terminal size $n = k^d - B - 1$ with $n = (M + \frac{1}{2})L + O(t^{d-1})$. The independent-fiber middle and eventually valid prefix yield $P_d(n) \geq P_d(M)^L$. Taking logarithms proves the bound. □

Theorem 16 (Power-divisor tensor recurrence)**.** *Fix an exponent $d \geq 2$ and a nonnegative integer n. For any odd positive integer t such that $L = t^d$ divides $2n + 1$, set*

$$A = \frac{L - 1}{2}, \qquad q = \frac{2n + 1}{L}, \qquad M_b = b^d - \frac{q + 1}{2}.$$

With $P_d(0) = 1$, we have

$$\boxed{P_d(n) \geq P_d(A) \sum_{\substack{b \geq 2,\ M_b \geq 0 \\ b^d < q}} P_d(M_b)^{\,L}.} \tag{18}$$

For $t = 1$, this is exactly the earlier one-step interval-reflection counting recurrence (now also allowing the empty prefix).

Proof. For each summand choose one permutation on $[1, A]$ and, independently, L permutations of $[1, M_b]$. Apply Lemma 14 on $[A+1, B]$, where $B = A + M_b L$. The equation defining M_b gives

$$B + n + 1 = b^d L = (bt)^d, \qquad n - B = (q - b^d)L > 0.$$

Thus the final interval $[B+1, n]$ is permuted by the reflection $x \mapsto (bt)^d - x$. All three pieces are valid and disjoint, producing $P_d(A)P_d(M_b)^L$ distinct permutations. Different b have different last entries $p(n) = (bt)^d - n$, so these families are disjoint and can be added. When $t = 1$, $L = 1$, $A = 0$ and $M_b = b^d - n - 1$, recovering the reflection recurrence. □

This formula gives a precise arithmetic description of the advantage and limitation of the tensor construction: a nontrivial block scale $t > 1$ requires the cubic divisibility condition $t^d \mid (2n+1)$. It does not, by itself, prove exponential growth at *every* length.

Proposition 4 (Arithmetic sparsity of nontrivial block lifting)**.** *The set of positive integers n for which some odd integer $t \geq 3$ satisfies $t^3 \mid (2n+1)$ has natural density*

$$\boxed{1 - \frac{8}{7\zeta(3)} = 0.0492487\ldots.}$$

Thus more than 95% of lengths have no nontrivial integral cube scale for the particular centered tensor construction used in Theorem 16.

Proof. An odd integer is cube-free precisely when it is divisible by no prime cube p^3, with p odd. By Möbius inclusion–exclusion, the number of positive $n \leq X$ for which $2n+1$ is cube-free equals

$$\begin{aligned}
\sum_{\substack{n \leq X \\ 2n+1 \text{ cube-free}}} 1 &= \sum_{\substack{d \leq (2X+1)^{1/3} \\ d \text{ odd}}} \mu(d) \#\{n \leq X : d^3 \mid 2n+1\} \\
&= X \sum_{\substack{d \geq 1 \\ d \text{ odd}}} \frac{\mu(d)}{d^3} + O(X^{1/3}) \\
&= X \prod_{p \text{ odd}} (1 - p^{-3}) + O(X^{1/3}) \\
&= \frac{8X}{7\zeta(3)} + O(X^{1/3}).
\end{aligned}$$

Every odd composite $t \geq 3$ has an odd prime divisor, so $t^3 \mid 2n+1$ is possible for some $t \geq 3$ exactly when $2n+1$ is not cube-free. Taking complements proves the claim. □

This is a limitation of the *specific centered tensor construction*, not an upper bound on $P(n)$ or a no-go theorem for other embeddings. In particular, the exponential-in-n subsequence result cannot by itself be promoted to a uniform exponential lower bound using the same integral scales for every n.

Lemma 15 (A denser finite alternating-cycle gadget)**.** *There exists a cube-sum permutation of $[1, 362]$ admitting at least 40 pairwise vertex-disjoint alternating cycles. Consequently,*

$$\boxed{P(362) \geq 2^{40}.}$$

Proof. An explicit certificate in `certificates/forty_cycles_362.json` describes a valid permutation using 43 integer interval-reflection rows $p(i) = c - i$ and lists 40 alternating directed cycles on 182 distinct vertices. Each listed edge is checked by the single integer identity $i + p(j) = k^3$. These cycles are vertex-disjoint and can be switched independently. The standalone `verify_362.py` checker directly verifies every row and edge without any matching solver or permanent algorithm. □

Theorem 17 (Improved explicit exponential subsequence)**.** *For every odd integer $t \geq 7$, let*

$$N_t = \frac{733t^3 - 1}{2}.$$

Then

$$\boxed{P(N_t) \geq P\left(\frac{t^3-1}{2}\right) 2^{40t^3} \geq 2^{40t^3}.}$$

In particular,

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \geq \frac{80\log 2}{733} > 0.0756.}$$

Proof. In Theorem 16 take $d = 3$, $M_b = 362$, $b = 9$. Then $b^3 = 729$ and $q = 2 \cdot 729 - 2 \cdot 362 - 1 = 733$, so $b^3 < q$ and $n = N_t$. At $t = 7$ the prefix length $A = (7^3 - 1)/2 = 171$ is a known positive index from the finite classification. For every larger odd t, $A \geq 364$, so the existence theorem applies. Hence $P(A) > 0$ and

$$P(N_t) \geq P(A)P(362)^{t^3} \geq 2^{40t^3}.$$

Since $N_t \sim (733/2)t^3$, division by N_t gives the stated exponential coefficient. □

The constant here is stronger than the earlier 236-seed coefficient $18\log 2/551$. It is a rigorously certified *lower bound*, not a claim of optimal asymptotic entropy.

This gives an *exponential limsup*, not an exponential lower bound for all n. In particular, it is logically distinct from the uniform, near-critical reflection estimates in Section 14.

13 Many permutations: an exponential cube-root bound

Besides qualitative multiplicity, the reflection family yields a strong quantitative statement. Let $K(n) = \lfloor \sqrt[3]{2n} \rfloor$, with the floor defined using integer comparisons (not floating point).

Lemma 16 (Three disjoint embeddings)**.** *If $K(n) = k \geq 27$, each of $k^3, (k-1)^3, (k-2)^3$ gives a reflection extension from a predecessor $m \geq 173$ with $m < n$, and all three predecessors obey $K(m) \geq k - 6$.*

Proof. By definition, $k^3 \leq 2n < (k+1)^3$. For the smallest of these cubes, set $m = (k-2)^3 - n - 1$. Then

$$2m > 2(k-2)^3 - (k+1)^3 - 2 = k^3 - 15k^2 + 21k - 19.$$

The right side exceeds $(k-6)^3$ for $k \geq 27$, since the difference is

$$3k^2 - 87k + 197 > 0 \quad (k \geq 27).$$

Thus $2m > (k-6)^3 \geq 21^3$, which implies $m \geq 173$ and $K(m) \geq k - 6$. Also $(k-2)^3 < k^3 \leq 2n$, so $m < n$. The other two cubes give larger predecessors and satisfy the same bounds. Distinct cubes yield disjoint matching families by Lemma 1. □

Theorem 18 (A096904 quantitative bound)**.** *For all integers $n \geq 173$, with $q(n) = \max\{0, \lfloor (K(n) - 27)/6 \rfloor\}$,*

$$\boxed{P(n) \geq 3^{q(n)}}.$$

In particular

$$\log P(n) \;\geq\; \frac{\log 3}{6}(2n)^{1/3} - O(1),$$

so $P(n)$ grows at least exponentially in $n^{1/3}$.

Proof. If $K(n) < 27$ then $q(n) = 0$, and Theorem 9 supplies $P(n) \geq 1$. For $K(n) \geq 27$, Lemma 16 provides three distinct extensions from smaller predecessors $m_i \geq 173$ with $K(m_i) \geq K(n) - 6$. Thus $q(m_i) \geq q(n) - 1$. By strong induction and the disjointness of the three extension images,

$$P(n) \geq \sum_{i=1}^{3} P(m_i) \geq 3 \cdot 3^{\max(0, q(n)-1)} \geq 3^{q(n)}.$$

The displayed logarithmic asymptotic follows immediately. □

14 A universal reflection-entropy bound

The fixed three-branch construction above is only one instance of a general branching mechanism. In fact, one may use *any fixed number* of the largest available cubes and obtain a sharp limiting coefficient for this family of lower bounds.

For each integer $s \geq 2$, let $\rho_s \in (1,2)$ be the unique solution of

$$\sum_{h=1}^{s} \rho_s^{-h} = 1, \qquad B_s = 2s^2 + 4s - 1. \tag{19}$$

Let $K(n) = \lfloor \sqrt[3]{2n} \rfloor$ as before.

Theorem 19 (Reflection entropy)**.** *For all $s \geq 2$ and $n \geq 173$,*

$$\boxed{P(n) \geq \rho_s^{(K(n)-B_s)/2}.} \tag{20}$$

In particular,

$$\boxed{\liminf_{n\to\infty} \frac{\log P(n)}{(2n)^{1/3}} \;\geq\; \frac{\log 2}{2}.} \tag{21}$$

Equivalently, for every $\varepsilon > 0$ there is a constant C_ε such that

$$P(n) \geq \exp\Big(\big(\tfrac{1}{2}\log 2 - \varepsilon\big)\,(2n)^{1/3} - C_\varepsilon\Big) \quad (n \geq 173).$$

Proof. Fix s and write $B = B_s$, $\rho = \rho_s$. We prove (20) by strong induction. For $K(n) < B$, the right side is at most 1, and $P(n) \geq 1$ by Theorem 9.

Now suppose $k = K(n) \geq B$. For each $j = 0, \ldots, s-1$, take the cube $c_j = (k-j)^3$ and put $m_j = c_j - n - 1$. Since $k^3 \leq 2n < (k+1)^3$, the following exact polynomial inequality holds:

$$\begin{aligned} 2m_j - (k-2j-2)^3 &\geq 2(k-j)^3 - (k+1)^3 - 1 - (k-2j-2)^3 \\ &= 3k^2 - (6j^2+24j+15)k \\ &\qquad + 6j^3 + 24j^2 + 24j + 6 \\ &\geq 0. \end{aligned}$$

Dividing the final polynomial by 3, its coefficient of $-k$ is $2j^2+8j+5$ and its constant term is positive. Hence the last inequality follows from $k \geq B \geq 2j^2+8j+5$. Also $k - 2j - 2 \geq k - 2s \geq 8$, so $m_j \geq 256 \geq 173$ and $K(m_j) \geq k - 2(j+1)$. Each $c_j \leq k^3 \leq 2n$, so $m_j < n$.

The s reflection extensions are pairwise disjoint by their distinct last images, as in Lemma 1. Using the strong induction hypothesis, we obtain

$$\begin{aligned} P(n) &\geq \sum_{j=0}^{s-1} P(m_j) \\ &\geq \sum_{j=0}^{s-1} \rho^{(K(m_j)-B)/2} \\ &\geq \rho^{(k-B)/2} \sum_{j=0}^{s-1} \rho^{-(j+1)} = \rho^{(k-B)/2}. \end{aligned}$$

This proves the first assertion. The ρ_s are strictly increasing and converge to 2, since their defining finite geometric sums approach $\sum_{h\geq 1} 2^{-h} = 1$. For each fixed s, B_s is a constant independent of n and $K(n) = (2n)^{1/3} + O(1)$. Therefore the ratio on the left side of (21) is at least $\frac{1}{2}\log\rho_s$ in the limit. Letting $s \to \infty$ proves (21). □

Corollary 7 (Other power sums, conditional on existence)**.** *For an integer $d \geq 2$, let $P_d(n)$ count permutations of $[1,n]$ in which every $i + p_i$ is a perfect dth power. If $P_d(n) > 0$ for all sufficiently large n, then*

$$\liminf_{n\to\infty} \frac{\log P_d(n)}{(2n)^{1/d}} \geq \frac{\log 2}{2}.$$

Proof. For any fixed s and $j = 0, \dots, s-1$, write $k = \lfloor (2n)^{1/d} \rfloor$ and use $c_j = (k-j)^d$, $m_j = c_j - n - 1$. The binomial theorem gives, uniformly over these finitely many j,

$$2(k-j)^d - (k+1)^d - (k-2j-2)^d = dk^{d-1} + O_{d,s}(k^{d-2}) > 1$$

for all sufficiently large k. It follows that m_j is a smaller positive length in the eventual-existence range and $\lfloor (2m_j)^{1/d} \rfloor \geq k - 2j - 2$. The distinct reflection families obey the same s-branch induction as Theorem 19, with a suitable constant initial offset in k. The ρ_s tend to 2 as before. □

Theorem 20 (A near-critical lower bound with explicit loss)**.** *For $K(n) = \lfloor \sqrt[3]{2n} \rfloor$ and every $n \geq 173$,*

$$\boxed{P(n) \geq \frac{127}{K(n)}\, 2^{(K(n)-127)/2}.} \tag{22}$$

More generally, for every real $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that

$$\boxed{P(n) \geq C_\varepsilon \frac{2^{K(n)/2}}{K(n)^\varepsilon} \qquad (n \geq 173).} \tag{23}$$

Proof. Write $k = K(n)$. The exact cubic polynomial inequality in the proof of Theorem 19 works whenever

$$k \geq B_s = 2s^2 + 4s - 1, \qquad s \geq 2;$$

it supplies s *disjoint* reflection embeddings with predecessor indices $m_h \in [173, n-1]$ and $K(m_h) \geq k - 2h$ for $h = 1, \dots, s$.

First set $F(k) = 2^{k/2}/k$, which is increasing for all integer $k \geq 3$. For $k \geq 128$ choose $s = \lfloor \log_2 k \rfloor$. Then $s \geq 7$ and $k \geq 2^s \geq B_s$: the last inequality holds at $s = 7$, since $128 \geq 125$, and is preserved upon doubling, because $2B_s \geq B_{s+1}$ for all $s \geq 7$. Moreover $k - 2s \geq 8$, so every predecessor is in the increasing range of F. By the reflection counting lemma,

$$\begin{aligned} \frac{\sum_{h=1}^s F(k-2h)}{F(k)} &= \sum_{h=1}^s \frac{2^{-h}k}{k-2h} \\ &\geq \sum_{h=1}^s 2^{-h}\left(1 + \frac{2h}{k}\right) \\ &\geq 1 - 2^{-s} + \frac{2}{k} > 1, \end{aligned}$$

using $\sum_{h=1}^s h2^{-h} \geq 1$ for $s \geq 2$ and $k < 2^{s+1}$. For $k < 128$, Theorem 9 ensures $P(n) \geq 1$. Because F is increasing for $k \geq 7$, this implies $P(n) \geq F(k)/F(127)$. Strong induction using the above inequality extends the same bound to all $k \geq 128$. Since $F(127) = 2^{127/2}/127$, this is exactly (22).

For the more general result, fix $\varepsilon > 0$ and define $F_\varepsilon(k) = 2^{k/2}/k^\varepsilon$. It is increasing for all sufficiently large k. For sufficiently large k, set $s = \lceil \log_2(k/\varepsilon) \rceil$, so $s = O_\varepsilon(\log k)$, $k \geq B_s$ and $2^{-s} \leq \varepsilon/k$. Each predecessor is in the increasing range of F_ε. By the elementary Bernoulli inequality,

$$\left(\frac{k}{k-2h}\right)^\varepsilon \geq 1 + \frac{2\varepsilon h}{k}.$$

Consequently

$$\begin{aligned} \frac{\sum_{h=1}^s F_\varepsilon(k-2h)}{F_\varepsilon(k)} &\geq 1 - 2^{-s} + \frac{2\varepsilon}{k}\sum_{h=1}^s h2^{-h} \\ &\geq 1 - 2^{-s} + \frac{2\varepsilon}{k} > 1. \end{aligned}$$

Choose an initial K_ε large enough for all these inequalities to hold at $k \geq K_\varepsilon$. By positivity, the bound $P(n) \geq C_\varepsilon F_\varepsilon(k)$ holds for $k < K_\varepsilon$ on taking $C_\varepsilon = 2^{-K_\varepsilon/2}$. The same strong induction proves it for every larger k, yielding (23). □

These bounds approach the *critical* reflection coefficient $\frac{1}{2}\log 2$ with a logarithmic (indeed arbitrarily small polynomial) loss. Their proofs use no new computer-assisted certificate beyond the existence theorem.

The coefficient $(\log 2)/2$ is a *proved lower bound*, not an asserted exact exponential growth rate. The argument needs only the qualitative existence Theorem 9; it makes no further use of numerical matching data.

15 An efficient constructive algorithm

The proof is also algorithmic. Given $n > 364$, choose the *least* cube $c = k^3 \geq n + 174$ and put $m = c - n - 1$. The cube-gap lemma ensures $173 \leq m < n$. Minimality gives

$$(k-1)^3 < n + 174 \leq k^3 \quad \Longrightarrow \quad 173 \leq m \leq 3k^2 - 3k + 173 = O(n^{2/3}).$$

Repeat until the remaining size is in the certified interval $173 \leq m \leq 364$, then fill each recorded interval by the explicit reflection $p_i = c - i$.

Proposition 5 (Linear-output construction)**.** *A valid cube-sum permutation of any $n \geq 173$ can be constructed in $O(n)$ unit-cost arithmetic operations, with $O(\log \log n)$ reflection layers.*

Proof. The predecessor bound $m = O(n^{2/3})$ implies $\log m \leq (2/3) \log n + O(1)$, so the number of iterations before the fixed-size base is $O(\log \log n)$. The returned intervals are disjoint and their union, together with the base interval, is $\{1, \ldots, n\}$. Therefore exactly n entries are written. Each exact integer cube root can be computed using $O(\log n)$ comparisons, which contributes $O(\log n \log \log n) = o(n)$ additional unit-cost operations. □

The accompanying `construct_linear.py` implements this procedure with exact integer cube roots, and can also display the recursion plan without allocating the full output.

16 Reproduction and limitations

All programs require only Python 3 standard-library modules:

```
python papers/A006063/verify.py
python papers/A006063/independent_match.py
python papers/A006063/verify_extended.py
python papers/A006063/construct_linear.py
python papers/A006063/verify_word_paths.py
python papers/A006063/count_236.py
```

The archived data consist of 109 core witnesses, the legacy 192-row all-base cross-check, 48 small positive permutations, 124 negative Hall sets, and compact pairs of alternating-cycle certificates. The infinite inductive and counting lemmas are mathematical arguments rather than extrapolations from numerical experiments.

The status is a *computer-assisted proof candidate* pending independent scrutiny of the finite certificate, its interpretation, and literature priority. The present manuscript does not compute the exact values of $P(n)$ in general, nor does it claim external peer review.

A All 191 short alternating-path words

The integer at the left is the new size n; the following one-digit symbols are the successive cube roots r_j in Lemma 9. Together with the explicit seed in Lemma 8, these words form an independently and locally checkable existence certificate for $173 \le n \le 364$.

174: 635555363635555363633555536
175: 6336
176: 65555556
177: 6364636
178: 6364625555556
179: 74635555555547
180: 74474247447
181: 6554746474556
182: 655556
183: 663647
184: 66
185: 66264746366
186: 66266
187: 7447
188: 7455665547
189: 666363647447447
190: 647
191: 6665547556
192: 75257
193: 7575555757
194: 66655666
195: 6264757
196: 663666626
197: 75746264757
198: 65575264757
199: 666
200: 66633666
201: 64747574757
202: 6474752556
203: 663466666
204: 757424757
205: 65755756
206: 6666446666
207: 666665546666
208: 64757
209: 6443756
210: 757464757
211: 6666357
212: 66664666
213: 757475757
214: 653756
215: 6666556
216: 7555666666665557
217: 75757375375757375757
218: 7555757573573757575557
219: 757455757464757
220: 75757554665575757
221: 757573753666375757
222: 7375757
223: 757455755557554757
224: 7363753637
225: 7557357447557
226: 75447557
227: 757537525735757
228: 7524665566666657
229: 746664667
230: 752757573637557
231: 75666637575725257
232: 757536357556666665757
233: 75666666557537
234: 7375757256666667
235: 77
236: 7466466466466466466475727574 7
237: 7375747
238: 74664747
239: 73444667
240: 7466466444667
241: 7337
242: 734664667
243: 73527537534667
244: 73667
245: 746666667
246: 766375375375266667
247: 7666666666666666667
248: 76666266667
249: 766662757475757266667
250: 747
251: 766665736666667
252: 747573375747
253: 7474466375747
254: 74744664664747
255: 74757377
256: 8
257: 76666667
258: 747525747
259: 85666658566277
260: 85662666666667
261: 7475375758
262: 8558
263: 7736666444758
264: 7737574744758
265: 8558534758
266: 766664758
267: 85585757377
268: 7477
269: 766666677477
270: 8575247477
271: 8575266667
272: 7475375375758575758
273: 855857573757447
274: 85266258
275: 7437447
276: 74586666666658
277: 868
278: 7447
279: 868525377
280: 773368
281: 76666477
282: 72755857477
283: 727
284: 777558
285: 727575868
286: 85475758
287: 76658
288: 7577666658
289: 8666857537
290: 8558
291: 856666667
292: 86468
293: 8558667
294: 737575727737
295: 86477737
296: 7775486477
297: 83747775868
298: 7577738
299: 7775468
300: 8646368585486868
301: 7368666468
302: 76644867
303: 734775557
304: 736868
305: 8677777
306: 7586666856768
307: 7684777558
308: 768458667
309: 77776667
310: 76848
311: 77
312: 836366625868
313: 775868577
314: 77457777
315: 7764777
316: 867738
317: 76664468
318: 7668667
319: 864866367
320: 764777
321: 8685485677
322: 7684847777
323: 75737668
324: 7667677367
325: 7586268
326: 7586847
327: 8685758268
328: 8626846668
329: 7577
330: 86856768667
331: 7585268368
332: 867
333: 777777
334: 7686855758
335: 76358668668
336: 784686847586868
337: 8638486777
338: 7468487
339: 77
340: 77583867
341: 7777577
342: 88288
343: 86388
344: 86867777778
345: 858557288
346: 85777488
347: 8586677768
348: 8855676668
349: 8848
350: 8485738686388
351: 86864485882778
352: 8686488
353: 8577664648
354: 85868684848
355: 877388
356: 84577488
357: 86777755778
358: 858557785588
359: 8468
360: 8586482775858
361: 8586855588
362: 883477778
363: 8528686488
364: 855557388

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